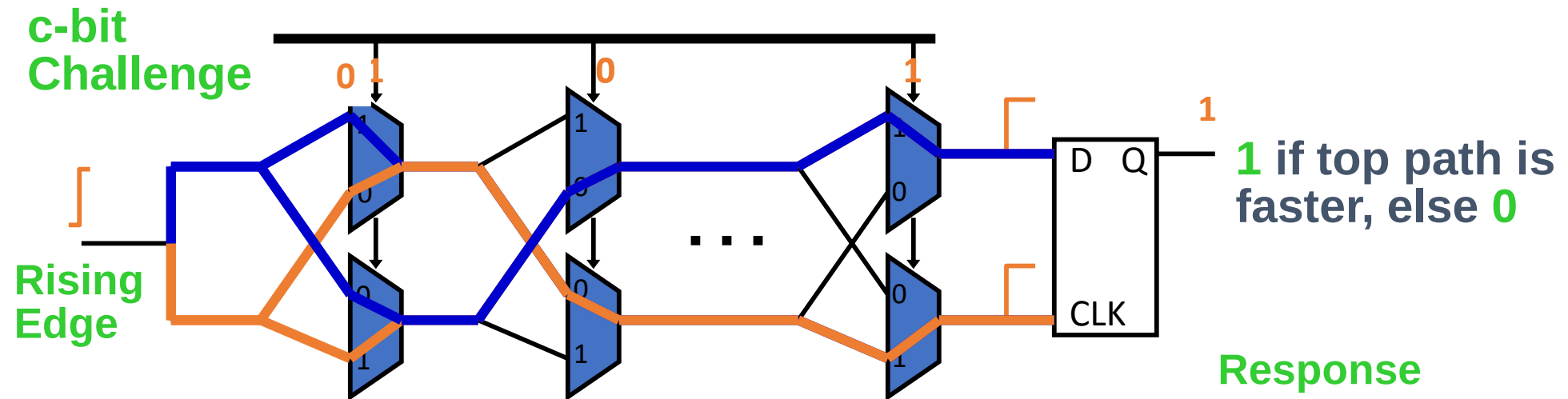


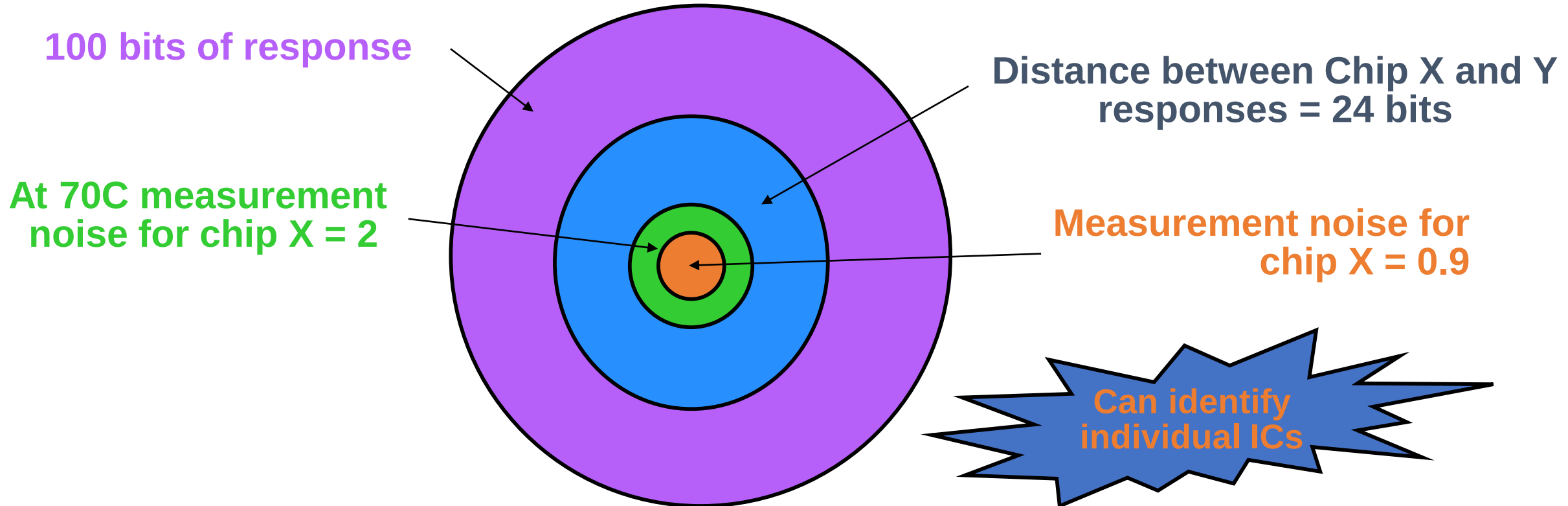
# Arbiter PUF



- Compare two paths with an identical delay in design
  - Random process variation determines which path is faster
  - An arbiter outputs 1-bit digital response
- Path delays in an IC are statistically distributed due to random manufacturing variations

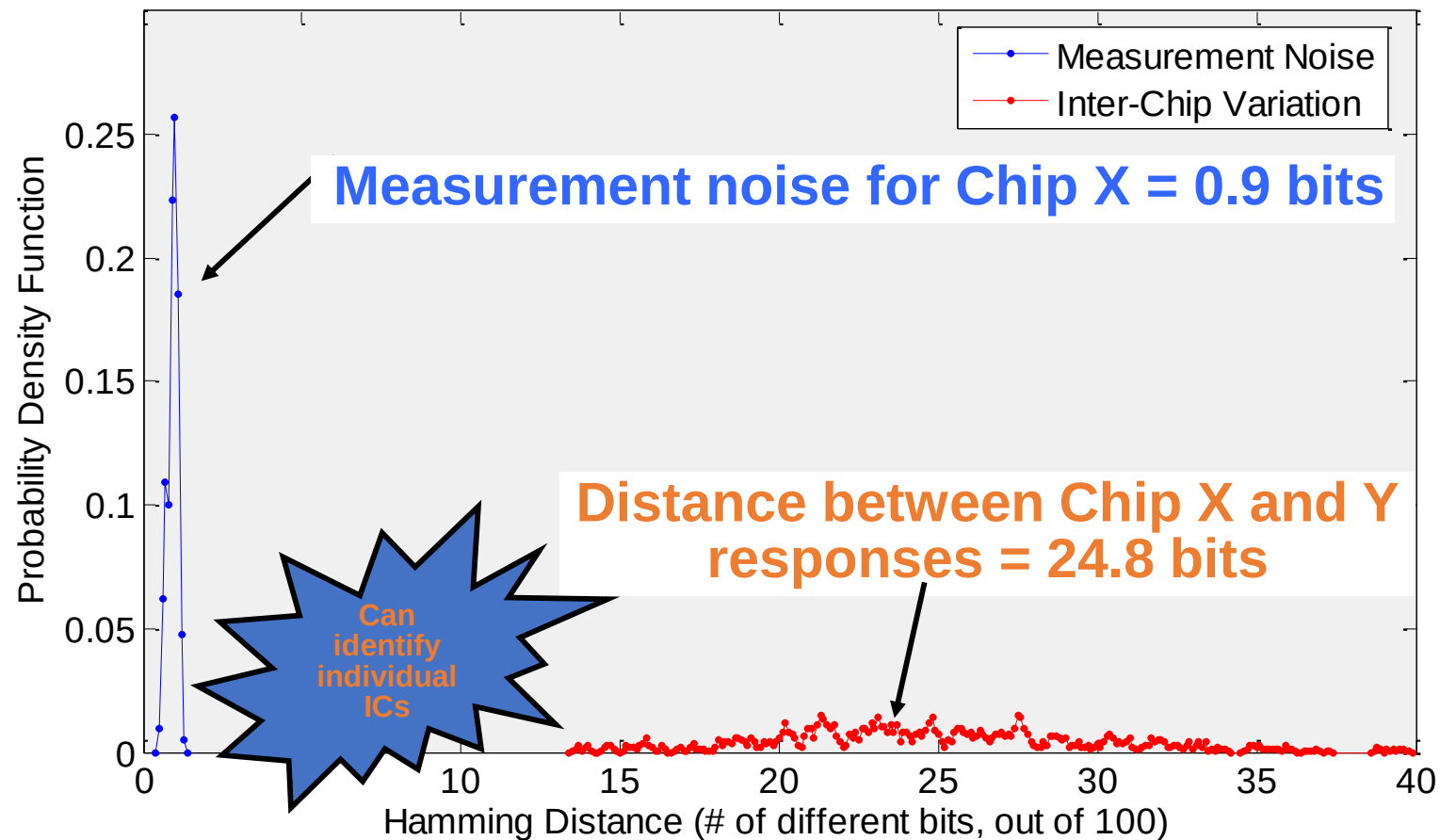
# Experiments

- Fabricated candidate PUF on multiple ICs, 0.18um TSMC
- Apply 100 random challenges and observe responses



# Inter-Chip Variation

- Apply random challenges and observe 100 response bits



# Environmental Variations

- What happens if we change voltage and temperature?

